

Architecture & Modeling Decisions Document

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1. Purpose

This document serves as the official architectural log and decision registry for the AdventureWorks Enterprise Sales Analytics Platform.

In enterprise data engineering, documenting the "Why" behind modeling decisions is just as critical as the execution itself. This document provides complete transparency regarding architectural trade-offs, logical design principles, data assumptions, and the strategic roadmap, ensuring alignment between business stakeholders and the technical implementation team.

2. Strategic Architectural & Modeling Decisions

Decision 01: Sales Domain as the Initial Data Product

- **Decision:** The initial release (Version 1.0) of the enterprise platform focuses exclusively on the Sales Domain, treating it as a standalone, high-value data product.
- **Scope Breakdown:**
 - **In-Scope Business Processes:** Internet Sales (B2C), Reseller Sales (B2B), and Sales Quota Planning (Sales Performance).
 - **Fact Tables Ingested:** FactInternetSales, FactResellerSales, and FactSalesQuota.
 - **Explicitly Excluded Domains:** Inventory, Finance, Manufacturing, Procurement, Human Resources, and Call Center.
- **Rationale:** Sales data constitutes the primary revenue engine of AdventureWorks and offers the most immediate strategic value to executive leadership. Concentrating on this domain ensures a rapid, high-quality deployment with minimized risk and logical clean-cut boundaries.
- **Expected Business Impact:** High executive visibility, simplified testing phases, and a stable foundation for incremental multi-domain integration.

Decision 02: Dimensional Modeling Methodology (Kimball Approach)

- **Decision:** Relational warehouse data will be modeled using the core principles of Ralph Kimball's **Dimensional Modeling Methodology**.
- **Rationale:** Unlike normalized operational databases (3NF), dimensional models are specifically optimized for rapid analytical queries, aggregate calculations, and user-friendly self-service reporting.
- **Expected Engineering Benefits:**
 - Reduction in query execution paths (minimizing joins).
 - Highly intuitive model structures for business analysts.
 - Native compatibility with Power BI's VertiPaq database engine.

Decision 03: Star Schema Architecture

- **Decision:** The physical and semantic layers will be structured strictly as a **Star Schema**, avoiding snowflaked dimensions wherever technically feasible.
- **Alternatives Evaluated:**

Alternative Design	Technical Evaluation	Decision Result
Snowflake Schema	Normalizes dimension tables (e.g., splitting Category and Subcategory). Increases complexity and join paths.	Rejected (Degrades DAX performance and reporting usability).
One Big Table (OBT)	Flattens everything into a single massive flat table. Leads to extreme redundancy and limits cross-dimensional analysis.	Rejected (Inflexible for multi-fact planning models like Quotas vs Sales).
Star Schema (Kimball)	Flattens dimension tables into single-level tables directly connected to centralized Fact tables.	Approved (Optimized for performance and simplicity).

- **Expected Engineering Benefits:** Sub-second visual rendering, simpler DAX logic, and an easily understandable drag-and-drop experience for self-service creators.

Decision 04: Single Unified Sales Semantic Layer

- **Decision:** Develop a **Single, Unified Semantic Dataset** in Power BI that houses all three transactional and planning processes (FactInternetSales, FactResellerSales, and FactSalesQuota), rather than creating siloed models.
- **Rationale:** B2C sales, B2B sales, and corporate quotas share key commercial dimensions (Date, Employee, Territory, Product). Combining them into a single managed dataset

guarantees standard KPI logic.

- **Expected Engineering Benefits:**
 - Eliminates redundant table loads.
 - Guarantees a single source of truth for cross-channel analysis.
 - Simplifies workspace access and report maintenance.

Decision 05: Mandatory Use of Conformed Dimensions

- **Decision:** Establish strictly defined **Conformed Dimensions** across the semantic layer.
- **Conformed Entities:** DimDate, DimProduct, DimSalesTerritory, and DimEmployee.
- **Rationale:** Conformed dimensions utilize identical surrogate keys and attributes across multiple fact tables. This allows users to seamlessly slice and compare online sales and wholesale sales using the same attributes (e.g., comparing internet and reseller sales within the same Calendar Year or Product Category).
- **Expected Business Impact:** Consistent filtering behavior, prevention of analytical discrepancies, and reliable cross-process business evaluations.

Decision 06: Order Date as the Active Date Relationship

- **Decision:** Link OrderDateKey to DimDate.DateKey as the **Active Relationship** in both transaction fact tables.
- **Role-Playing Fields Available:** OrderDateKey, ShipDateKey, and DueDateKey.
- **Rationale:** The vast majority of strategic business questions focus on when a transactional sale was committed (ordered), rather than when it was shipped or delivered.
- **Modeling Treatment:**
 - **Active Path:** OrderDateKey $\rightarrow$ DimDate (Default calendar filtering).
 - **Inactive Paths:** ShipDateKey and DueDateKey linked to DimDate. These will remain dormant and will be programmatically activated in DAX using USERELATIONSHIP() only when performing logistics or fulfillment variance analysis.

Decision 07: Power BI VertiPaq Import Storage Mode

- **Decision:** Ingest all tables into Power BI utilizing **Import Mode** instead of DirectQuery.
- **Rationale:** The current size of the AdventureWorksDW database comfortably fits within Power BI's in-memory thresholds. Import mode leverages the columnar VertiPaq compression engine, delivering unrivaled calculation performance.
- **Expected Engineering Benefits:** Immediate dashboard rendering, full access to advanced DAX time-intelligence functions, and reduced transactional query load on the production SQL Server.

Decision 08: Single-Direction Cross-Filtering Directionality

- **Decision:** Restrict all dimension-to-fact relationships to a **Single Filter Direction** ($1 \rightarrow *$).
- **Rationale:** Enabling bi-directional relationships introduces ambiguity, performance

overhead, and the risk of unexpected data-filtering behavior (filter bleeding).

Single-direction filtering ensures the logical data model behaves predictably.

- **Expected Engineering Benefits:** Clean query execution paths, reduced model processing times, and simplified model debugging.

Decision 09: Decoupled and Dedicated Measure Layer

- **Decision:** Implement all business calculations strictly as **DAX Measures** inside a non-physical virtual folder (e.g., _KPI Measures), completely avoiding the use of calculated columns in fact tables.
- **Rationale:** Calculated columns are evaluated during data refresh and stored statically, consuming valuable RAM and storage space. DAX measures are computed dynamically at query time, keeping the model lightweight and fast.
- **Expected Engineering Benefits:** Reduced model footprint, flexible runtime evaluations based on user filters, and structured organization of business logic.

Decision 10: Centralized KPI Definition Standards

- **Decision:** Standardize all high-level calculations (Revenue, Costs, Margins, Quota Attainments) globally in the semantic model.
- **Rationale:** To prevent "metric drift" (where different regional units calculate margins or profit differently), report developers must only consume pre-validated measures from the shared semantic dataset.
- **Expected Business Impact:** Higher data trust, standardized management reviews, and unified enterprise performance reporting.

Decision 11: Role-Oriented Presentation Strategy (Power BI Apps)

- **Decision:** Segment report delivery into **Three Specialized Power BI Applications** tailored to specific organizational personas.



- **App 1: Executive Overview App (Strategic):** High-level aggregate KPIs, corporate growth, global margins, and macro trends for C-Level leadership.
- **App 2: Sales Performance App (Operational):** Representative quota achievements, territory performance tables, and regional target tracking for Sales Managers.
- **App 3: Product & Customer App (Tactical):** Detailed SKU velocity, customer demographic profiling, and promotion ROI matrices for Category Managers.
- **Rationale:** Delivering a single massive dashboard to all users leads to information overload. Role-based applications ensure stakeholders only interact with relevant data.

Decision 12: Version-Based Expansion Strategy

- **Decision:** Implement a progressive development lifecycle, establishing Sales as the core foundation before horizontally scaling to other domains.
- **Engineering Roadmap:**
 - **Version 1.0 (Current):** Dual-Channel Sales Performance & Employee Quota Target Tracking.
 - **Version 2.0 (Planned):** Product Inventory Integration (turnover ratios, safety stock levels).
 - **Version 3.0 (Planned):** Financial Ledger Integration (Direct General Ledger reconciliation).
- **Rationale:** Progressive delivery mitigates project risk, shortens the time-to-market for the initial product, and ensures high quality before increasing complexity.

3. Core Business & Technical Assumptions

- **Assumption 01 (Data Integrity):** The underlying database tables in AdventureWorksDW are assumed to be complete, clean, and representative of the true transactional history. No complex data-cleansing pipelines are expected at the Power Query level.
- **Assumption 02 (Key Uniqueness):** All surrogate keys generated in dimension tables (e.g., ProductKey, CustomerKey) uniquely identify historical dimension members and are stable over time.
- **Assumption 03 (Transaction Accuracy):** The transactional records in FactInternetSales and FactResellerSales accurately reflect net sales transactions and correspond directly to the corporate ERP records.
- **Assumption 04 (Quota Validity):** The targets loaded into FactSalesQuota represent the finalized, audited, and approved sales quotas assigned to representatives by corporate leadership.
- **Assumption 05 (Currency Standardization):** Financial revenue calculations in local currencies are mapped correctly in the warehouse, and the primary reporting currency is standardized on USD.
- **Assumption 06 (Contiguous Calendar):** The conformed dimension DimDate contains a continuous, unbroken series of calendar days matching the historical date boundaries of

our sales facts, which is essential for DAX time-intelligence functions.

- **Assumption 07 (Analytical Focus):** Version 1.0 prioritizes high-level corporate analytical capabilities (trends, averages, aggregates, and goals) over operational, real-time transaction monitoring.

4. Future Architectural Enhancements

Upon successful delivery of Version 1.0, the platform's technical architecture is prepared to scale horizontally to support the following enhancements:

1. **Dynamic Row-Level Security (RLS):** RESTRICT data access so that Sales Representatives can only view performance metrics belonging to their assigned Sales Territories.
2. **Microsoft Fabric Migration (Direct Lake Mode):** Move the storage layer into Microsoft Fabric (OneLake SaaS), running calculations directly from Delta tables to bypass traditional cloud scheduled refreshes.
3. **Advanced Customer Lifetime Value (LTV):** Introduce predictive machine learning models to forecast future customer purchase behavior and segmentation.